

Nash Bargaining for Wholesaler Territory Division

A discrete formulation over postal codes

Abstract

Two merging wholesaling forces must divide a common territory. We formulate the problem directly over postal codes and solve it by Nash bargaining from the zero disagreement point: neither wholesaler can unilaterally revert to a pre-merger book once the merger is executed, so the outside option is nothing, not the legacy territory. Ordering zips by the ratio of the two wholesalers' utilities gives a fast heuristic and, via outer approximation on the concave log-objective, an exact solution with an optimality certificate. Nash is Pareto efficient by construction, satisfies envy-freeness up to one zip, and cannot purchase fairness by making both parties worse off — a failure mode the equalisation criteria do exhibit. Contiguity is then imposed exactly by a districting integer program — solved by outer approximation, since the log of the Nash product is concave rather than linear — and costs under half a percent of the bargaining product. A fifty-zip instance is worked through in full: allocation, contiguity, contestability triage, and exact parameter breakpoints. Alternative solution concepts and the constructions specific to them are collected in the appendices.

1 Setup

Index postal codes $z = 1, \dots, n$, each carrying pre-merger sales A_z to wholesaler a and B_z to wholesaler b , and total market opportunity M_z (demand net of what either has already written); write $M = \sum_z M_z$. Two parameters govern how a zip's value splits between whichever wholesaler is assigned it and whichever is not. *Transfer capture* $\theta \in [0, 1]$ is the share of a book the non-incumbent retains after taking over the zip — attrition and persistency account for $1 - \theta$. *Headroom credit* $\lambda \in [0, 1]$ is how much a dollar of unwritten opportunity counts against a dollar of booked production.

Headroom is taken *net*: whoever is assigned z is credited with opportunity net of business already on the books there, not gross of it. Gross double-counts and is incoherent whenever $\lambda > \theta$ — it would make an inherited book of business a *liability*, since the same dollar of production is worth more credited as headroom than credited as a captured book. Under the net convention, with

$$c_1 = 1 - \lambda, \quad c_2 = \theta(1 - \lambda), \quad (1)$$

the value of zip z to each wholesaler, were it assigned the whole territory, is

$$u_a(z) = c_1 A_z + c_2 B_z + \lambda M_z, \quad u_b(z) = c_2 A_z + c_1 B_z + \lambda M_z. \quad (2)$$

u_a credits wholesaler a with all of their own book, a θ -share of b 's, and a λ -share of what neither has written; u_b is the mirror image. Net headroom is feasible only if it is non-negative pointwise,

$$M_z \geq \max(A_z + \theta B_z, B_z + \theta A_z) \quad \text{for every } z, \quad (3)$$

which we take as maintained throughout. An *allocation* is a partition of $Z = \{1, \dots, n\}$ into a set S given to wholesaler a and its complement $Z \setminus S$ given to b : every zip is assigned, to exactly one side, with no third disposition.

2 The bargaining problem

This section poses territory division as a two-person bargaining problem in the classical sense (Nash 1950), then works down from that object to the program actually solved. Seven layers.

2.1 Layer 1 — players, outcomes, and the feasible set

The players are the two wholesalers, a and b . An outcome is an allocation $S \subseteq Z$ as defined above, giving each player the payoff

$$U_a(S) = \sum_{z \in S} u_a(z), \quad U_b(S) = \sum_{z \notin S} u_b(z), \quad (4)$$

using the utilities (2). The *feasible set*

$$F = \{(U_a(S), U_b(S)) : S \subseteq Z\} \subset \mathbb{R}^2 \quad (5)$$

is the set of attainable payoff pairs: a cloud of at most 2^n points, entirely determined by the data (A_z, B_z, M_z) and the parameters θ, λ .

2.2 Layer 2 — the disagreement point

A bargaining problem is a pair (F, d) , where $d \in \mathbb{R}^2$ is the *disagreement point*: the payoff each player secures unilaterally, without the other's cooperation, if no allocation is agreed. Getting d right is not a normalisation — it is a claim about what each wholesaler can actually walk away with, and it is the single most consequential modelling choice in this paper.

An earlier draft set $d = (S_a, S_b)$, each wholesaler's pre-merger production: if the division fails, each keeps the book they walked in with. This is the natural first guess, and it is wrong. A disagreement point must be an outcome a player can execute *unilaterally* if bargaining breaks down. Once the merger closes, pre-merger production is not such an outcome: the separate sales forces, appointments, and client relationships that generated S_a and S_b are dissolved into the merged entity, and neither wholesaler can unilaterally resurrect them regardless of how the territory dispute resolves. S_a is a historical number, not a threat point. What each wholesaler can secure by failing to agree is nothing: absent an assignment, neither is appointed or staffed to write business anywhere in the disputed territory. The disagreement point is therefore the origin,

$$d = (0, 0). \quad (6)$$

This is not a matter of taste. $d = (S_a, S_b)$ with $S_a \neq S_b$ is not a neutral rescaling: bargaining from an asymmetric threat point is observationally equivalent to an *asymmetric* Nash solution at $d = 0$ with an implicit, undisclosed weight (here $\omega \approx 0.54$ on the worked instance) — a tilt toward whichever side had the bigger pre-merger book, smuggled in through the disagreement point rather than argued for on its merits. It also forfeits envy-freeness up to one good (EF1): the Nash bargaining solution is EF1 at $d = 0$ (Caragiannis et al. 2019) and the guarantee does not survive moving d away from the origin. Both defects disappear at $d = (0, 0)$.

Writing $g_a(S) = U_a(S) - 0$ and $g_b(S) = U_b(S) - 0$ for the gains relative to d , gains are simply bundle utilities:

$$g_a(S) = \sum_{z \in S} u_a(z), \quad g_b(S) = \sum_{z \notin S} u_b(z), \quad G_a^{\max} = \sum_z u_a(z), \quad G_b^{\max} = \sum_z u_b(z), \quad (7)$$

with $G_a^{\max}, G_b^{\max}$ the attainable maxima from receiving the whole territory. Because $\theta, \lambda \in [0, 1]$ and (3) holds, $u_a(z), u_b(z) \geq 0$ pointwise, so $g_a(S) > 0$ for every non-empty S and $g_b(S) > 0$ for every $S \neq Z$: unlike the pre-merger baseline, where headroom below a threshold λ^* left the bargaining set empty, feasibility of both gains is automatic here for any non-trivial split. That threshold was an artefact of the discarded convention and does not arise at $d = 0$.

2.3 Layer 3 — Nash’s axioms and the solution

Nash (1950) shows that if F is convex and compact and contains a point strictly dominating d , exactly one point of F satisfies four axioms:

Pareto optimality. No other feasible point makes both players weakly better off and one strictly better off.

Symmetry. If F is symmetric in the two players and $d_a = d_b$, the solution gives them equal payoff.

Invariance to affine transformations. The solution commutes with independently rescaling and shifting either player’s utility — utility is cardinal only up to a positive affine transformation, and the solution must not depend on the choice of units.

Independence of irrelevant alternatives. Shrinking F to a subset that still contains the solution does not change it.

That point is the maximiser of the *Nash product*, the product of gains over the disagreement point:

$$\max_{(U_a, U_b) \in F} (U_a - d_a)(U_b - d_b) \quad \text{subject to } U_a > d_a, U_b > d_b. \quad (8)$$

No other point of F satisfies all four axioms simultaneously; this is what makes (8) a characterisation rather than one criterion among several equally defensible ones.

F is not convex here. Nash’s theorem is stated for convex, compact F ; with indivisible postal codes F is a finite point cloud instead, so strictly the axioms select a *lottery* over two adjacent allocations, and (8) maximised over the point cloud is the discrete analogue rather than the axiomatic solution itself. The gap is quantified in the remark on convexity in Section 3: 0.0034% on the worked instance. Known, small, ignored in practice; worth re-checking on real data.

2.4 Layer 4 — specializing to the discrete allocation problem

At $d = (0, 0)$, applying (8) to the feasible set generated by allocations gives the criterion actually solved throughout this paper:

$$\max_{S \subseteq Z} g_a(S) g_b(S), \quad g_a(S) > 0, g_b(S) > 0, \quad (9)$$

with g_a, g_b as in (7). The remainder of this section turns (9) into a program that can be handed to a solver: first restricting F by operational constraints (contiguity, state borders), then writing the result as a mixed-integer program, then solving it exactly.

2.5 Layer 5 — constraints shrink F

Operational requirements restrict which S are admissible, and therefore which points of F are attainable. Contiguity — each wholesaler receiving one connected territory on the postal-code adjacency graph $G = (Z, E)$ — is the binding one:

$$F_{\text{cont}} = \{(g_a(S), g_b(S)) : G[S] \text{ and } G[Z \setminus S] \text{ both connected}\} \subseteq F. \quad (10)$$

The same bargaining problem is then solved on the smaller set. *That is the precise game-theoretic content of the districting constraints: they do not change the criterion, they shrink the feasible set.* State-border restrictions act identically, by deleting cross-state edges from G before connectivity is assessed.

2.6 Layer 6 — the program

Maximising (8) over F_{cont} at $d = (0, 0)$, with $x_z = \mathbf{1}[z \in S]$:

$$\max_{x \in \{0,1\}^n} \log g_a + \log g_b \quad (11)$$

$$\text{s.t. } g_a = \sum_z x_z u_a(z), \quad g_b = \sum_z (1 - x_z) u_b(z), \quad (12)$$

$$g_a > 0, \quad g_b > 0, \quad (13)$$

$$G[S] \text{ and } G[Z \setminus S] \text{ connected.} \quad (14)$$

This is a convex MINLP. The gains (12) are *linear* in x ; the objective (11) is *concave* in those linear expressions; connectivity (14) is representable by linear constraints. Maximising a concave function over a mixed-integer linear feasible set is the well-behaved MINLP class, for which outer approximation is a finite global method. The product form $\max g_a g_b$ is *not* concave — it is bilinear and indefinite — so the logarithm is not cosmetic: it is what places the problem in the tractable class.

2.7 Two representations of connectivity

Constraint (14) is not a linear inequality as written. Two standard encodings.

(a) Compact, single-commodity flow. Fix roots r_a, r_b . Introduce $f_{uv}^a \geq 0$ on each arc. Every zip assigned to a absorbs one unit of flow sent from r_a through arcs internal to a 's territory:

$$\sum_{(u,z) \in E} f_{uz}^a - \sum_{(z,v) \in E} f_{zv}^a = x_z, \quad z \neq r_a, \quad (15)$$

$$f_{uv}^a \leq (n-1)x_u, \quad f_{uv}^a \leq (n-1)x_v, \quad (u,v) \in E, \quad (16)$$

with the mirror system in $(1-x)$ for b . Polynomial size, $O(|E|)$ continuous variables per district, one solve.

(b) Exponential, separator cuts. For every $S' \subseteq Z$ inducing a component not containing the root,

$$|S'| \sum_{w \in N(S')} x_w \geq \sum_{v \in S'} x_v. \quad (17)$$

There are exponentially many such inequalities, so they are generated *lazily*: solve, inspect the incumbent for disconnected components, add the violated cuts, re-solve. In practice a handful of rounds suffice. This is the encoding used here; it keeps the model small at the cost of repeated solves.

2.8 Layer 7 — solving it

Outer approximation linearises the concave objective from above. Introduce $z_a \leq \log g_a$, $z_b \leq \log g_b$ enforced by supporting hyperplanes (19), and maximise $z_a + z_b$. Each relaxation is a pure MILP; its optimum is an upper bound on (11); a tangent added at the incumbent tightens it. Interleaving the tangent cuts with the connectivity cuts (17) converges both simultaneously, and termination when the incumbent’s true log-product meets the bound is an optimality *certificate*. Compactness may be priced by subtracting $\rho \sum_e y_e$ with $y_e \geq |x_u - x_v|$, at the cost of changing the objective.

layer	object	form
1	players, outcomes, feasible set F	2^n points
2	disagreement point	$d = (0, 0)$
3	Nash’s axioms $\Rightarrow$ solution	$\max(U_a - d_a)(U_b - d_b)$
4	discrete specialisation	$\max g_a(S)g_b(S)$
5	constraints (contiguity, state borders)	shrink F
6	program	(11)–(14), convex MINLP
7	method	outer approximation + lazy cuts, finite, certified

3 The Nash solution

Section 2 reduced fair division to the criterion (9), $\max_{S \subseteq Z} g_a(S)g_b(S)$ subject to $g_a, g_b > 0$. This section gives the exact algorithm. A fast, solver-free heuristic based on ordering zips by exchange rate — useful as a preview or sanity check, but not exact for Nash — is developed in Appendix B.

Taking logarithms, $\log(g_a g_b) = \log g_a + \log g_b$, does not linearise the objective: g_a is linear in x , so $\log g_a$ is *concave* in a linear expression. The problem is therefore a convex MINLP — maximising a concave function over a MILP-representable set — for which outer approximation is a finite *global* method. Introduce $z_a \leq \log g_a$ and add supporting hyperplanes

$$z_a \leq \log(\hat{g}) + \frac{g_a - \hat{g}}{\hat{g}}, \quad (18)$$

one linear cut per tangent point, no additional binaries, and maximise $z_a + z_b$. The tangent set is a relaxation, so the MILP optimum is a valid *upper bound*; adding a tangent at the incumbent and re-solving tightens it, and when the incumbent’s true log-product meets the bound the incumbent is globally optimal. This is a certificate, not a stopping heuristic.

Validated against exhaustive enumeration, 120 random instances per size: exact match in 100% of cases at every n tested, with bound gap identically zero. Cost is 6–7 iterations regardless of size — 0.10s at $n = 50$, 0.84s at $n = 400$. **Use the exact method by default**; the prefix table (Appendix B) is a fast preview and a sanity check.

Remark (Convexity, and what "Nash" means over indivisible zips). Nash's axiomatic characterisation — Pareto efficiency, symmetry, scale invariance, independence of irrelevant alternatives — assumes the bargaining set is *convex*. With indivisible postal codes the feasible set $F = \{(g_a(S), g_b(S))\}$ is a finite point cloud, so strictly the axioms deliver a *lottery* over two adjacent allocations, and $\max g_a g_b$ over F is the natural discrete analogue rather than the axiomatic solution itself. The distinction is quantitatively small here: the Nash product on the convex hull, where randomisation is permitted, is 24.09177 against 24.09096 for the best deterministic allocation — a difference of 0.0034%. The frontier is dense enough that the deterministic vertex lies essentially on the hull. It is worth re-checking on real data, and not worth randomising territory assignments over.

Remark (No fixed point is needed). In a continuum the Nash threshold satisfies $t^* = g_a/g_b$, a fixed point whose naive iteration diverges — on this instance it runs $1.000 \rightarrow 40.22 \rightarrow -0.244 \rightarrow -6.447$ into a degenerate corner. Neither the prefix table nor the outer approximation requires it.

4 Why Nash

Exactly solvable. Outer approximation (Section 3) reaches a certified global optimum in 6–7 iterations regardless of n . The equalisation criteria require search for their exact optima and carry no such certificate.

Pareto efficient by construction. Maximising a product cannot be improved for both parties simultaneously. The min-|difference| criteria carry no such guarantee and do fail in practice: an unconstrained equalising program reached a gap of 0.000161 while giving up 18% of attainable welfare (Appendix A).

Envy-freeness up to one zip. Neither wholesaler prefers the other's territory once a single zip is removed from it — the strongest fairness guarantee available for indivisible goods, and legible in a room.

The one limitation. Nash requires $g_a, g_b > 0$, so below the feasibility threshold λ^* the bargaining set is empty. Report λ^* explicitly; under the net convention it was 0.069 on the instances tested, well below any defensible λ . Below it the merger is negative-sum for the wholesalers and the conversation is not about criteria. The egalitarian criterion is the fallback there.

If seniority should count. Asymmetric Nash, $\max g_a^\omega g_b^{1-\omega}$, retains the prefix property with threshold $\frac{1-\omega}{\omega} \cdot \frac{g_a}{g_b}$. An explicit weight is preferable to a criterion that encodes a tilt nobody selected.

5 A fifty-zip instance

Fifty postal codes on the unit square drawn from a two-component Gaussian mixture, with metros at (0.28, 0.62) and (0.72, 0.36); a is anchored on the first, b on the second, both present in each. Totals $S_a = 3.0$, $S_b = 1.8$, $M = 40$ (12% combined penetration); $\theta = 0.4$, $\lambda = 0.3$.

$\text{corr}(A_z, B_z)$ across zips	+0.685
utility ratio range	[0.9531, 1.1373]
minimum net headroom	0.2335

The positive correlation is the harder configuration: both wholesalers are strong in the same places. The ratio spans a narrow band mostly above 1, so on raw utility a is favoured nearly everywhere and fairness is the only force conceding territory to b .

50 zips on the unit square — Voronoi cells coloured by field; marker size $\propto$ magnitude; stars mark the two metro centres

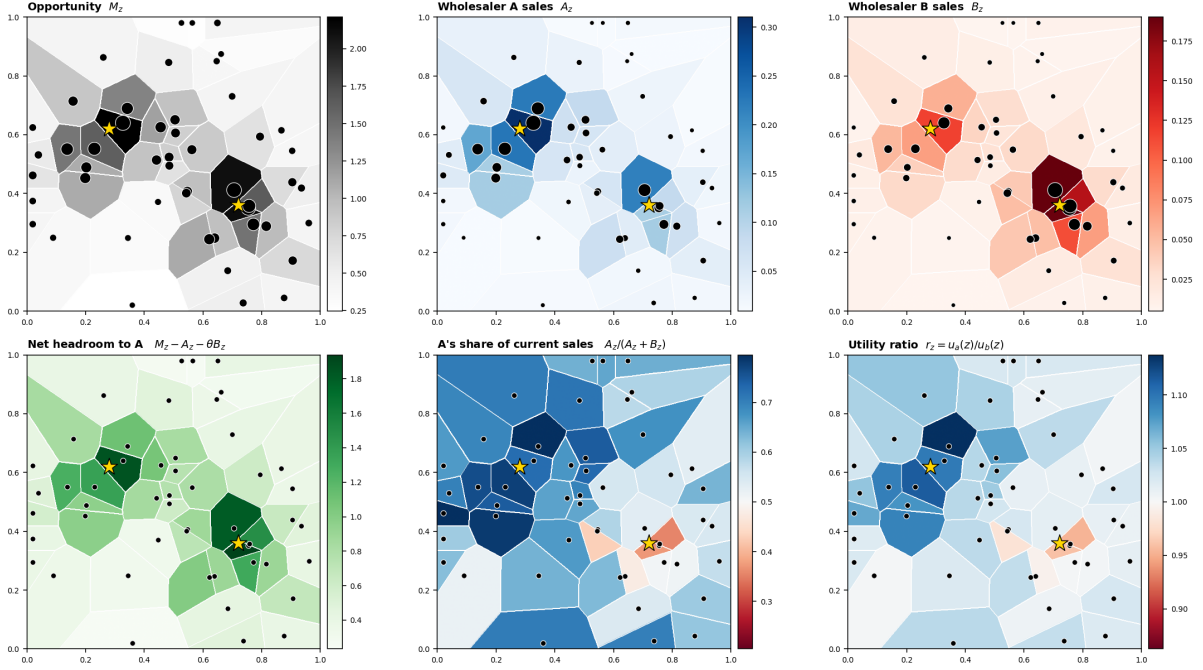


Figure 1: The instance before optimisation. Voronoi cells coloured by field, marker size proportional to magnitude, stars at metro centres.

5.1 Solution

	k	g_a	g_b	$g_a g_b$
Nash, exact (outer approximation)	25	4.9787	4.8388	24.09117
Nash, prefix heuristic (Appendix B)	25	4.9360	4.8807	24.09096

The two select *different* allocations of the same size; the heuristic's shortfall is 8.4×10^{-6} , which is negligible here but not zero.

a receives 25 zips, 53.1% of opportunity. Equal gain and the egalitarian criterion select the same allocation; Kalai–Smorodinsky differs by one zip (Appendix A).

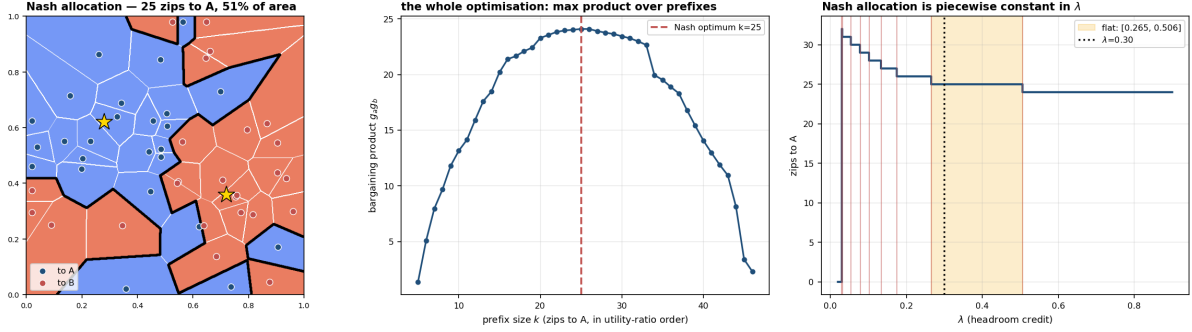


Figure 2: Left: the Nash allocation. Centre: the entire optimisation — the bargaining product over the $n + 1$ prefixes, maximised at $k = 25$. Right: the allocation is piecewise constant in λ , with $\lambda = 0.30$ inside a flat interval running from 0.265 to 0.506.

6 Contiguity

Nothing above forces each territory to be connected, and the Nash allocation is not: a holds two pieces and b three, in each case a stray single zip. Imposing connectivity turns the problem into a *districting* integer program — the formulation used for electoral redistricting and sales-territory alignment — with one binary per zip.

6.1 Formulation: the log makes it concave, not linear

Taking logarithms separates the Nash product, $\log(g_a g_b) = \log g_a + \log g_b$, but does not linearise it: g_a is linear in x , so $\log g_a$ is *concave* in a linear expression. That is the useful case. A concave function is the lower envelope of its tangents, so introduce $z_a \leq \log g_a$ and add supporting cuts

$$z_a \leq \log(\hat{g}) + \frac{g_a - \hat{g}}{\hat{g}}, \quad (19)$$

one linear constraint per tangent point $\hat{g}$, with no additional binaries. The program is then

$$\text{decision } x_z \in \{0, 1\}, \text{ 1 if } z \rightarrow a, \quad (20)$$

$$\text{objective } \max z_a + z_b - \rho \sum_{e \in E} y_e, \quad y_e \geq |x_u - x_v|, \quad (21)$$

with z_a, z_b bounded by tangent cuts (19). The tangent set is an *outer* approximation — an upper bound — which tightens as tangents are added at the incumbent solution. This is Kelley’s cutting-plane method, and it interleaves naturally with the contiguity cuts of Section 6.2: both converge together, typically in about five iterations.

The y_e count *boundary edges*, the perimeter of the partition, and ρ prices compactness. Note that $\rho > 0$ *changes the objective*: the solve is then exact for Nash-minus-a-perimeter-penalty, a legitimate modelling choice but not exact Nash. Set $\rho = 0$ for the latter.

Two traps. First, the compactness term is not optional: fairness alone is degenerate, many allocations achieve near-identical objective values, and contiguity cuts never bite — an implementation without it thrashed for seventeen rounds without converging. Second, a weighted-sum scalarisation $w g_a + (1 - w) g_b$ collapses to corner solutions when the frontier is close to linear, as it

is here, and an ε -constraint sweep needs one MILP solve per frontier point. Outer approximation avoids both, at roughly a twentieth of the cost.

6.2 Contiguity by separator cuts

Fix a root zip per side. If a 's territory splits, then for each component S not containing a 's root add

$$|S| \sum_{w \in N(S)} x_w \geq \sum_{v \in S} x_v, \quad (22)$$

“if any of S is with a , some neighbour just outside must be too”; symmetric in $(1 - x)$ for b . Re-solve and repeat. The alternative is a single-commodity *flow* formulation — ship one unit from each assigned zip to its root along in-district edges — exact in one solve but requiring $O(|E|)$ extra continuous variables per district. Aggregating one cut per island rather than one per zip matters for convergence.

6.3 Result

The exact unconstrained Nash allocation leaves a in three pieces and b in two, in each case a stray single zip. Enforcing connectivity, and sweeping the compactness weight on a graph of 50 zips and 132 adjacency edges:

ρ	$g_a g_b$	perimeter	% of edges	iterations	time	pieces a/b
— (unconstrained)	24.0910	—	—	—	—	2/3
2×10^{-3}	24.0146	15	11.4%	5	0.7s	1/1
2×10^{-4}	24.0569	18	13.6%	5	0.1s	1/1
1×10^{-5}	24.0774	33	25.0%	10	0.4s	1/1

The price of contiguity under Nash is between 0.06% and 0.32% of the bargaining product, depending on how clean a boundary is wanted. Contiguity is close to free when solved exactly. A greedy island-repair heuristic, by contrast, degraded the objective by more than three orders of magnitude and should be used only to *detect* fragmentation, never to fix it.

ρ competes directly with the log objective and should be swept rather than fixed: at $\rho = 10^{-5}$ the boundary cuts a quarter of all adjacency edges for a 0.26% gain in product over the compact solution. Report the product and the perimeter together.

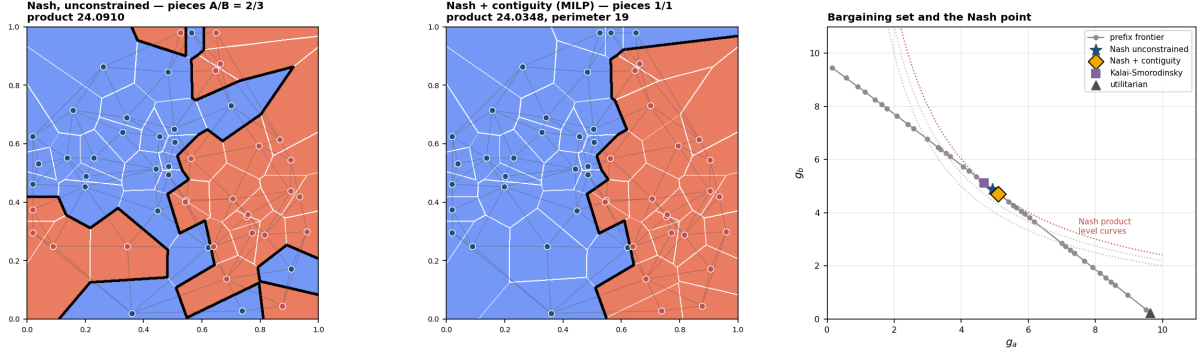


Figure 3: Left: unconstrained Nash with the Rook adjacency overlaid; each wholesaler holds a stray single-zip island. Centre: the contiguous MILP solution. Right: the bargaining set — grey points are the prefix frontier, dotted curves are level sets of $g_a g_b$, and the Nash point is where the highest attainable level curve touches the frontier. The utilitarian corner leaves b with 0.24.

7 Contestability

Where are the hard decisions? Two components, multiplied.

Stake. Misassigning zip z moves the bargaining product by, to first order,

$$\text{stake}_z = \frac{u_a(z)}{g_a} + \frac{u_b(z)}{g_b}, \quad (23)$$

the relative swing in $\log g_a g_b$. On the instance it ranged over $[0.0336, 0.3556]$.

Doubt. The flip rate $2 \min(p, 1-p)$ under uncertainty. Sweeping $\theta \in [0.20, 0.60]$ and $\lambda \in [0.10, 0.50]$ moves few zips; perturbing the *data* by 10% on sales and 6% on opportunity moves 34 of 50.

Parameter uncertainty is nearly irrelevant here; data uncertainty dominates. Getting A_z and B_z right matters more than settling λ .

rank	zip	stake	doubt	quadrant
1	32	0.3387	0.362	adjudicate
2	22	0.1378	0.792	adjudicate
3	19	0.1028	0.873	adjudicate
4	28	0.0882	0.915	coin-flip
5	10	0.0597	0.992	coin-flip

Four zips fall in the adjudicate quadrant; the top five carry 52% of total contestability. Zip 32 carries by far the largest stake and moderate doubt, so it is both the biggest and the most genuinely open question — unlike under an equalising criterion, where it registered as large but settled.

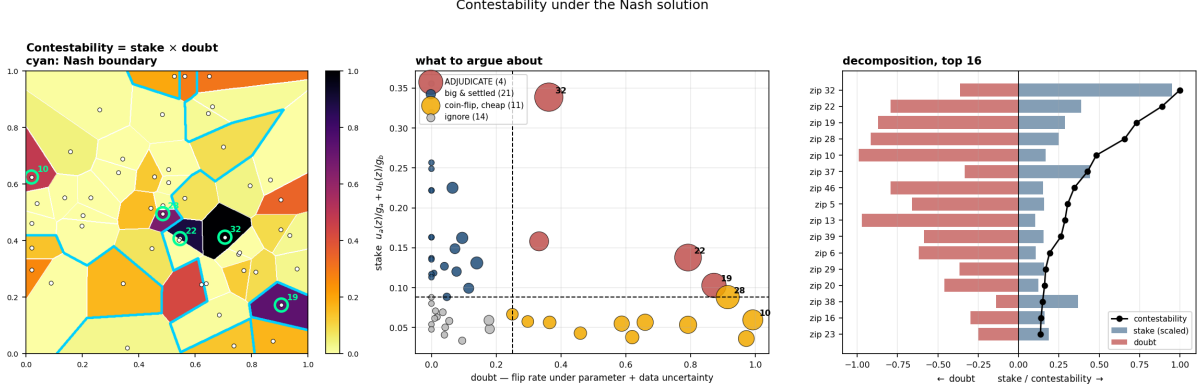


Figure 4: Contestability as stake $\times$ doubt. Left: mapped, Nash boundary in cyan, top five ringed. Centre: the 2×2 . Right: decomposition of the top sixteen.

8 Exact parameter breakpoints

Write $P_z = A_z + \theta B_z$, $Q_z = B_z + \theta A_z$, so $u_a(z) = (1 - \lambda)P_z + \lambda M_z$. Zips i, j exchange places in the ratio order exactly when

$$(1 - \lambda)K_{ij} + \lambda L_{ij} = 0 \implies \lambda_{ij}^* = \frac{K_{ij}}{K_{ij} - L_{ij}}, \quad \begin{aligned} K_{ij} &= P_i Q_j - P_j Q_i, \\ L_{ij} &= P_i M_j - P_j M_i + Q_j M_i - Q_i M_j. \end{aligned} \quad (24)$$

One explicit expression per pair; in θ the condition is quadratic, hence also closed form. On the instance the Nash allocation takes only **ten distinct values** across $\lambda \in [0.02, 0.90]$.

$\lambda = 0.30$ sits inside a flat interval from 0.265 to 0.506: anyone who believes a dollar of untapped opportunity is worth between roughly 27 and 50 cents of booked production obtains the same map. Nearly all sensitivity is crowded below $\lambda \approx 0.14$.

This is a stronger instrument than a local derivative: rather than reporting $dk/d\lambda$, one states the complete set of possible answers and exactly which belief produces each.

9 Opportunity balance

Before the merger both wholesalers covered the whole market, each with access to $M = 40$ at penetrations of 7.5% and 4.5%. The honest framing is that both end with roughly half of what they had. Under Nash, $M_a = 21.23$ (53.1%) against $M_b = 18.77$, an imbalance of 6.1%.

λ is the opportunity-balance dial. The headroom term places most of the weight on opportunity at moderate λ , so the criterion balances market size without being asked to. Imbalance falls monotonically from 19.5% at $\lambda = 0.02$ to about 1% at $\lambda = 0.70$. “How much should untapped opportunity count?” and “should territories carry equal market size?” are the same question.

Imposing equal opportunity as a hard condition is possible but is a strong position: with books of 3.0 against 1.8 it fully discounts the larger book. That is a compensation-philosophy decision rather

than a modelling one.

10 Implementation and scope

`territory.py` operates on a `networkx` graph of ZCTAs with Rook adjacency and attributes `rep_a`, `rep_b`, `A`, `B`, `M`, `state`: it validates headroom, builds the bipartite overlap graph of wholesaler pairs and reports its component structure, solves each pair by prefix table, and writes results back for mapping. `districting.py` adds the contiguous MILP, with `solve_contiguous_nash` performing the ε -constraint sweep. Setting `respect_state=True` deletes cross-state edges before enforcing connectivity; for life and annuity distribution this may bind harder than contiguity, since product availability and producer appointments are state-scoped.

Not covered. Capacity and travel constraints. Three or more wholesalers, for which no threshold rule exists and the natural extension is leximin over each dense component of the overlap graph. Estimation of θ from historical reassignments, on which everything ultimately rests. At national scale, a solver with lazy-constraint callbacks is preferable to re-solving from scratch each round.

Three checks on real data.

1. $\text{corr}(A_z, B_z)$ across zips — whether the books overlap or separate, and hence how contested the exercise is.
2. Connected components of the allocation on the true adjacency graph — pricing contiguity rather than assuming it.
3. Measurement error on A_z, B_z, M_z — since data uncertainty, not parameter uncertainty, sets which zips need adjudication.

A Other solution concepts

Selecting each criterion directly over the fifty-one prefixes of the instance in Section 5:

crit ^{erion}	k	g_a	g_b	$g_a g_b$	$\min(g)$	$ g_a/G_a^{\max} - g_b/G_b^{\max} $
Nash	25	4.9360	4.8807	24.091	4.8807	0.0286
equal gain	25	4.9360	4.8807	24.091	4.8807	0.0286
egalitarian	25	4.9360	4.8807	24.091	4.8807	0.0286
Kalai–Smorodinsky	24	4.6807	5.1301	24.012	4.6807	0.0137
equal opportunity	22	4.4423	5.3628	23.823	4.4423	0.0532
utilitarian	46	9.6330	0.2395	2.307	0.2395	0.8107

Nash, equal gain and the egalitarian criterion coincide. Kalai–Smorodinsky differs substantively rather than by artefact: it equalises *fractions* of attainable gain, and $G_b^{\max} = 12.30$ exceeds $G_a^{\max} = 11.60$ precisely because b brought the smaller book and has the lower baseline to clear. KS therefore awards more absolute gain to the wholesaler who brought less production — defensible in principle, awkward to deliver. The utilitarian row shows what fairness buys: unconstrained, a takes 46 zips and leaves b a gain of 0.24.

Equalisation can destroy value. Minimising a difference is not a monotone objective, and the unconstrained equalising MILP illustrates the hazard: it reached a KS gap of 0.000161 at 82.1% of attainable welfare, against 85.8% for the best prefix. The remedy is a welfare floor $g_a + g_b \geq \kappa \max_S(g_a + g_b)$, whose right-hand side is available in closed form because the utilitarian optimum is the prefix $\{z : u_a(z) > u_b(z)\}$. Nash needs no such guard.

B The prefix heuristic

Section 3 solves (9) exactly by outer approximation. This appendix records a fast, solver-free alternative — a sanity check and a preview, not a substitute — and the extent to which it can be trusted.

B.1 Definition, and exactness for the utilitarian criterion only

Define the utility ratio

$$r_z = \frac{u_a(z)}{u_b(z)}, \quad (25)$$

the rate at which utility transfers between the wholesalers when zip z changes hands.

Proposition 1 (Prefix heuristic). *Order the zips so that $r_{(1)} \geq \dots \geq r_{(n)}$ and write $P_k = \{(1), \dots, (k)\}$. Then the utilitarian optimum is exactly P_k with $k = |\{z : r_z > 1\}|$, and $\arg \max_k g_a[k]g_b[k]$ is a fast approximation to the Nash optimum.*

The utilitarian claim is exact: the objective is $\sum_{z \in S} (u_a(z) - u_b(z)) + \text{const}$, maximised by taking every zip with $u_a > u_b$. The Nash claim is not. The usual argument — at an optimum every included zip satisfies $u_a(z)/g_a \geq u_b(z)/g_b$, a threshold on r_z — is *marginal*, treating g_a, g_b as fixed while a zip moves. Discretely they jump by a finite amount, so the threshold characterisation is only asymptotically valid.

The prefix rule is an approximation, and it is wrong about half the time. Against exhaustive enumeration over 200 random instances per size:

n	prefix optimal	mean shortfall	worst shortfall
6	40.0%	3.91%	53.9%
10	47.5%	1.17%	13.9%
14	45.0%	0.57%	10.7%
n	shortfall vs. local optimum		
20	0.16%		
50	0.02%		
100	0.002%		
400	0.0009%		

The error vanishes as the discretisation refines, because the jumps shrink relative to the gains. At realistic sizes it is far below the uncertainty in θ or in the sales data — but it is not zero,

and the rule should be described as a heuristic, never substituted for Section 3’s exact method where a certificate matters.

B.2 Where it fails on other criteria

Proposition 1 covers Nash and the utilitarian criterion. It does not extend to criteria that minimise a *distance*, for which an off-frontier subset can hit the target more precisely. Over 400,000 random subsets of the worked instance:

criterion	best prefix	best random subset	
utilitarian $\max g_a + g_b$	32.983	32.894	prefix holds
Nash $\max g_a g_b$	210.253	201.896	prefix holds
egalitarian $\max \min(g)$	13.382	14.096	violated
equal gain $\min g_a - g_b $	2.3287	0.0001	violated
Kalai–Smorodinsky	0.02240	0.000004	violated

The prefix table therefore remains exact for Nash, and is a fast restriction for the others — one with the merit of staying on a family of allocations ordered by exchange rate.

C The mixture-quantile shortcut

For the *equalisation* criteria the continuum analysis yields an appealing shortcut. Collecting (7) under $g_a = g_b$ gives

$$\Phi(S) = q, \quad \Phi(S) = \frac{\Lambda \sum_{z \in S} (A_z + B_z) + 2\lambda \sum_{z \in S} M_z}{D}, \quad q = \frac{S_a(1 + c_2) - \lambda S_b + \lambda M}{D}, \quad (26)$$

with $\Lambda = c_1 + c_2$ and $D = \Lambda(S_a + S_b) + 2\lambda M$: accumulate mixture mass along the ratio order and cut where it crosses q . Kalai–Smorodinsky takes the same form with weights $K_a = S_a(G_b^{\max} c_1 + G_a^{\max} c_2)$, $K_b = S_b(G_b^{\max} c_2 + G_a^{\max} c_1)$, $K_M = \lambda M(G_a^{\max} + G_b^{\max})$ and level $q_{KS} = (G_a^{\max} G_b^{\max} + G_b^{\max} S_a)/K$.

On a continuum this is exact. On discrete data it is off by one zip: q is a continuum target while the cumulative mass *jumps*, so truncation need not give the nearer prefix. With $q_{KS} = 0.523326$ the crossing falls between $k = 24$ (mass 0.517624) and $k = 25$ (0.535206). On the equal-gain criterion the same truncation yields a gap of 0.449 where $k = 25$ gives 0.055.

The mixture form remains valuable as *theory* — it makes q interpretable, yields closed-form comparative statics, and reveals that λ governs the weight on opportunity (Section 9) — but it is the wrong algorithm for discrete data, and it does not arise for Nash at all.